
Least Squares

Annex A — Two Derivations of One Equation

an impossible system · minimising the squared error · projecting onto $\text{col}(\mathbf{X})$ · why
these are the same thing

Computing, Robotics & Bionics

3rd-year Bioengineering — companion to A. Géron, *Hands-On Machine Learning with
Scikit-Learn and PyTorch*

*This annex expands §14.1 of Refresher Module 2. Linear regression is introduced there twice — once as the minimisation of a squared error, once as an orthogonal projection — and the two accounts are asserted to agree. Here they are derived separately, from first principles, and then shown to be the same n equations written in two languages. The document is **self-contained**: it assumes only the scalar product, norms and projection of Module 2, §6, and the partial derivative of Module 3, Part 1. It closes with the rank-deficient case, which is where the two viewpoints stop agreeing about the answer and start disagreeing usefully.*

Prerequisites: Module 2 (§§1–6, 14) and Module 3 Part 1 (§§1–3). Companion notebook:
[m2_eigen_svd_pca](#).

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1 An impossible system

Suppose we have m measurements and want to explain each one as a linear combination of n features. Collect the features in a matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ — one row per sample, one column per feature — and the measured responses in $\mathbf{y} \in \mathbb{R}^m$. We are looking for a weight vector $\mathbf{w} \in \mathbb{R}^n$ such that

$$\mathbf{X}\mathbf{w} = \mathbf{y}. \quad (1.1)$$

Written out, (1.1) is a system of m linear equations in n unknowns. In any realistic setting $m \gg n$: hundreds of patients and a dozen measurements, thousands of recordings and sixteen acoustic features. The system is **overdetermined**, and overdetermined systems generically have no solution at all.

The algebraic statement of this is the theorem you met before university, usually under the name **Rouché–Frobenius**: the system $\mathbf{X}\mathbf{w} = \mathbf{y}$ is consistent if and only if

$$\text{rank}(\mathbf{X}) = \text{rank}([\mathbf{X} \mid \mathbf{y}]), \quad (1.2)$$

that is, appending $\mathbf{y}$ as an extra column must not increase the rank. With $m > n$ the matrix $\mathbf{X}$ has rank at most n , while the augmented matrix has $n + 1$ columns and can have rank $n + 1$. Any measurement noise whatsoever — and there is always measurement noise — will push it there.

Key idea

The geometry behind the rank condition. Condition (1.2) says exactly that $\mathbf{y}$ is a linear combination of the columns of $\mathbf{X}$; equivalently, $\mathbf{y} \in \text{col}(\mathbf{X})$.

Now count dimensions. The column space $\text{col}(\mathbf{X})$ is a subspace of $\mathbb{R}^m$ of dimension at most n . The data vector $\mathbf{y}$ lives in the whole of $\mathbb{R}^m$. When $m = 5875$ and $n = 19$, we are asking whether a point in 5875-dimensional space happens to land exactly inside a 19-dimensional slice of it. It does not, and it never will.

So (1.1) is not a system to be solved. It is a system to be *approximated*.

A biosensor calibration that does not fit

A biosensor is calibrated against four standards of known concentration $c = 0, 1, 2, 3$, giving readings $y = 1, 3, 4, 8$. We want a straight line $y = w_0 + w_1c$, so

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 8 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} w_0 \\ w_1 \end{pmatrix}.$$

Four equations, two unknowns. The first two, $w_0 = 1$ and $w_0 + w_1 = 3$, already force $\mathbf{w} = (1, 2)$; the third then demands $1 + 4 = 4$, which is false. Here $\text{rank}(\mathbf{X}) = 2$ but $\text{rank}([\mathbf{X} \mid \mathbf{y}]) = 3$, so by (1.2) the system is inconsistent. No straight line passes through all four points.

This is not a defect of the experiment. It is the normal situation, and it is why every regression routine you will ever call solves a different problem from the one in (1.1).

2 Choosing what to minimise

If we cannot make the residual

$$\mathbf{r}(\mathbf{w}) = \mathbf{y} - \mathbf{X}\mathbf{w} \quad (2.1)$$

vanish, we can at least make it small. But “small” needs a definition, and Module 2, §6.3 established that a vector space admits many: the L_1 norm $\sum_i |r_i|$, the L_2 norm $\sqrt{\sum_i r_i^2}$, the L_∞ norm $\max_i |r_i|$, and infinitely many others. Each gives a different “best” line.

We choose L_2 , and minimise its square:

Definition

The **least-squares problem** is

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^n} S(\mathbf{w}), \quad S(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 = \sum_{i=1}^m (y_i - \mathbf{x}_i^\top \mathbf{w})^2, \quad (2.2)$$

where $\mathbf{x}_i^\top$ denotes the i -th row of $\mathbf{X}$, i.e. the features of sample i . Squaring the norm changes nothing about where the minimum is — $t \mapsto t^2$ is increasing on $t \geq 0$ — and removes an awkward square root.

Four reasons for this choice, in increasing order of depth.

It is differentiable. S is a polynomial in the entries of $\mathbf{w}$, so we can differentiate it and set the derivative to zero. The L_1 objective has a kink wherever a residual crosses zero, and L_∞ has kinks everywhere.

It has a closed form. We are about to derive it. L_1 and L_∞ regression are perfectly respectable — they are linear programs, and L_1 is more robust to outliers, being a conditional-median rather than a conditional-mean estimator — but neither has a formula.

It has a statistical pedigree. If the measurement errors are independent and Gaussian with constant variance, the least-squares estimate is the maximum-likelihood estimate; the Gauss–Markov theorem further says it has the smallest variance among all unbiased linear estimators.

And — the reason this annex exists — L_2 is the only one of the three that comes from a scalar product. Module 2, §6.3 tested this with the parallelogram law: L_1 and L_∞ fail it, L_2 passes. That is not a curiosity. Angles, orthogonality and projection are defined by a scalar product, not by a norm. So L_2 is the *only* choice for which “the best fit” will turn out to have a geometric meaning at all.

3 First derivation: set the gradient to zero

Expand (2.2) using $\|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v}$:

$$\begin{aligned} S(\mathbf{w}) &= (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) \\ &= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w} \\ &= \mathbf{y}^\top \mathbf{y} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}. \end{aligned} \quad (3.1)$$

The two middle terms were combined because each is a 1×1 matrix — a scalar — and a scalar equals its own transpose, so $\mathbf{y}^\top \mathbf{X}\mathbf{w} = (\mathbf{y}^\top \mathbf{X}\mathbf{w})^\top = \mathbf{w}^\top \mathbf{X}^\top \mathbf{y}$.

Now differentiate with respect to $\mathbf{w}$. Two standard gradients do all the work: for constant $\mathbf{a}$ and symmetric $\mathbf{A}$,

$$\nabla_{\mathbf{w}}(\mathbf{w}^\top \mathbf{a}) = \mathbf{a}, \quad \nabla_{\mathbf{w}}(\mathbf{w}^\top \mathbf{A}\mathbf{w}) = 2\mathbf{A}\mathbf{w}. \quad (3.2)$$

Here $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$, which is symmetric because $(\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top \mathbf{X}$. Applying (3.2) to (3.1),

$$\nabla S(\mathbf{w}) = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{w}. \quad (3.3)$$

Setting this to $\mathbf{0}$ and dividing by two gives the result.

The normal equations

Any minimiser $\hat{\mathbf{w}}$ of S satisfies

$$\mathbf{X}^\top \mathbf{X} \hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y} \quad (3.4)$$

and, when $\mathbf{X}^\top \mathbf{X}$ is invertible,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (3.5)$$

3.1 Why the stationary point is a minimum

Setting a gradient to zero locates a stationary point; it does not by itself tell you whether you have found a minimum, a maximum or a saddle. Differentiate (3.3) once more:

$$\nabla^2 S = 2 \mathbf{X}^\top \mathbf{X}. \quad (3.6)$$

The Hessian is constant — S is exactly quadratic, with no higher terms — and it is positive semi-definite, because for any $\mathbf{v} \in \mathbb{R}^n$

$$\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{v} = (\mathbf{X} \mathbf{v})^\top (\mathbf{X} \mathbf{v}) = \|\mathbf{X} \mathbf{v}\|^2 \geq 0. \quad (3.7)$$

A function with a positive semi-definite Hessian everywhere is convex, and for a convex function every stationary point is a *global* minimum. There are no local minima to get stuck in and no saddle points to escape — which is why linear regression, alone among the models in this course, needs no iterative optimiser.

Uniqueness is a slightly stronger requirement. If $\text{rank}(\mathbf{X}) = n$ — the columns are linearly independent, no feature is a combination of the others — then $\mathbf{X} \mathbf{v} = \mathbf{0}$ only for $\mathbf{v} = \mathbf{0}$, so (3.7) is strictly positive for $\mathbf{v} \neq \mathbf{0}$, the Hessian is positive definite, S is strictly convex, and the minimiser is unique. We return in Section 9 to what happens when it is not.

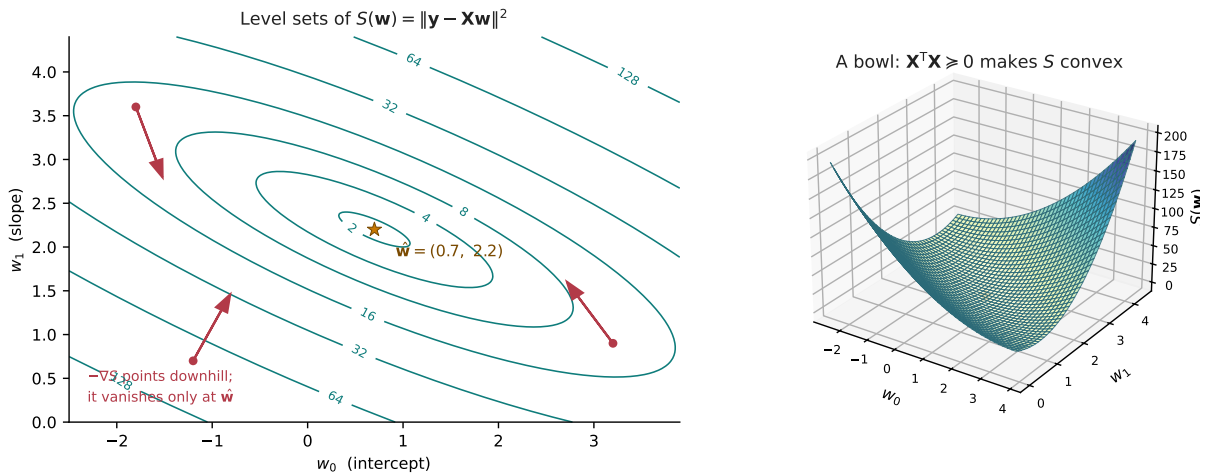


Figure 1: The error surface for the calibration example. Because $\mathbf{X}^\top \mathbf{X}$ is positive definite, S is a strictly convex paraboloid with elliptical level sets and a single lowest point. The negative gradient points downhill from anywhere, and vanishes only at $\hat{\mathbf{w}} = (0.7, 2.2)$, where $S = 1.80$.

3.2 The same thing, one component at a time

Before moving on, write (3.3) out for a single weight. Let $\mathbf{x}^{(j)}$ denote the j -th *column* of $\mathbf{X}$, so that $\mathbf{X} \mathbf{w} = \sum_j w_j \mathbf{x}^{(j)}$. Differentiating the sum form of (2.2) directly by the chain rule,

$$\frac{\partial S}{\partial w_j} = \sum_{i=1}^m 2(y_i - \mathbf{x}_i^\top \mathbf{w}) \cdot (-X_{ij}) = -2 \sum_{i=1}^m X_{ij} r_i = -2 \langle \mathbf{x}^{(j)}, \mathbf{r} \rangle. \quad (3.8)$$

Keep (3.8) in view. It is the whole of Section 5.

4 Second derivation: find the closest point in a subspace

Now forget calculus entirely.

As $\mathbf{w}$ ranges over all of $\mathbb{R}^n$, the product $\mathbf{X}\mathbf{w}$ ranges over all linear combinations of the columns of $\mathbf{X}$ — that is, over $\text{col}(\mathbf{X})$, and nothing else. So (2.2) is not really a question about $\mathbf{w}$ at all. It is this question:

Which point of the subspace $\text{col}(\mathbf{X})$ is closest to $\mathbf{y}$?

Once that point is found, $\mathbf{w}$ is merely its list of coordinates in terms of the columns. This is a question about a point and a subspace, and it has a clean answer that predates regression by a long way.

Closest point in a subspace

Let $V \subseteq \mathbb{R}^m$ be a subspace and $\mathbf{y} \in \mathbb{R}^m$. A vector $\hat{\mathbf{y}} \in V$ minimises $\|\mathbf{y} - \mathbf{v}\|$ over $\mathbf{v} \in V$ if and only if the residual is orthogonal to the whole subspace:

$$\mathbf{y} - \hat{\mathbf{y}} \perp V, \quad \text{that is} \quad \langle \mathbf{y} - \hat{\mathbf{y}}, \mathbf{v} \rangle = 0 \quad \text{for every } \mathbf{v} \in V. \quad (4.1)$$

Such a $\hat{\mathbf{y}}$ exists and is unique; it is the **orthogonal projection** of $\mathbf{y}$ onto V .

Proof. Suppose $\hat{\mathbf{y}} \in V$ satisfies (4.1), and let $\mathbf{v} \in V$ be any other candidate. Split the error into two pieces:

$$\mathbf{y} - \mathbf{v} = \underbrace{(\mathbf{y} - \hat{\mathbf{y}})}_{\perp V} + \underbrace{(\hat{\mathbf{y}} - \mathbf{v})}_{\in V}.$$

The second piece lies in V because V is a subspace and both terms belong to it; the first is orthogonal to everything in V by assumption. The two are therefore orthogonal to each other, and Pythagoras applies:

$$\|\mathbf{y} - \mathbf{v}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{v}\|^2 \geq \|\mathbf{y} - \hat{\mathbf{y}}\|^2, \quad (4.2)$$

with equality exactly when $\|\hat{\mathbf{y}} - \mathbf{v}\| = 0$, i.e. $\mathbf{v} = \hat{\mathbf{y}}$. So $\hat{\mathbf{y}}$ beats every alternative strictly, which proves both minimality and uniqueness. $\square$

Where the scalar product was used

Equation (4.2) is the only step of the proof that does any work, and it is available only because the norm comes from a scalar product. Repeat the argument in L_1 and it collapses at that line: there is no Pythagoras, no orthogonality, and the closest point in a subspace need not even be unique.

4.1 Applying it to regression

Take $V = \text{col}(\mathbf{X})$ and $\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{w}}$. Condition (4.1) requires the residual to be orthogonal to every vector in $\text{col}(\mathbf{X})$. It is enough to check the n columns that span it, since orthogonality to a spanning set extends to all combinations by linearity:

$$\langle \mathbf{x}^{(j)}, \mathbf{y} - \mathbf{X}\hat{\mathbf{w}} \rangle = 0 \quad \text{for } j = 1, \dots, n. \quad (4.3)$$

Those n scalar equations are the n rows of a single matrix equation — stacking the columns $\mathbf{x}^{(j)}$ as the rows of $\mathbf{X}^\top$ is precisely what transposition does:

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\mathbf{w}}) = \mathbf{0} \iff \mathbf{X}^\top\mathbf{X}\hat{\mathbf{w}} = \mathbf{X}^\top\mathbf{y}. \quad (4.4)$$

Which is (3.4) again, obtained without differentiating anything.

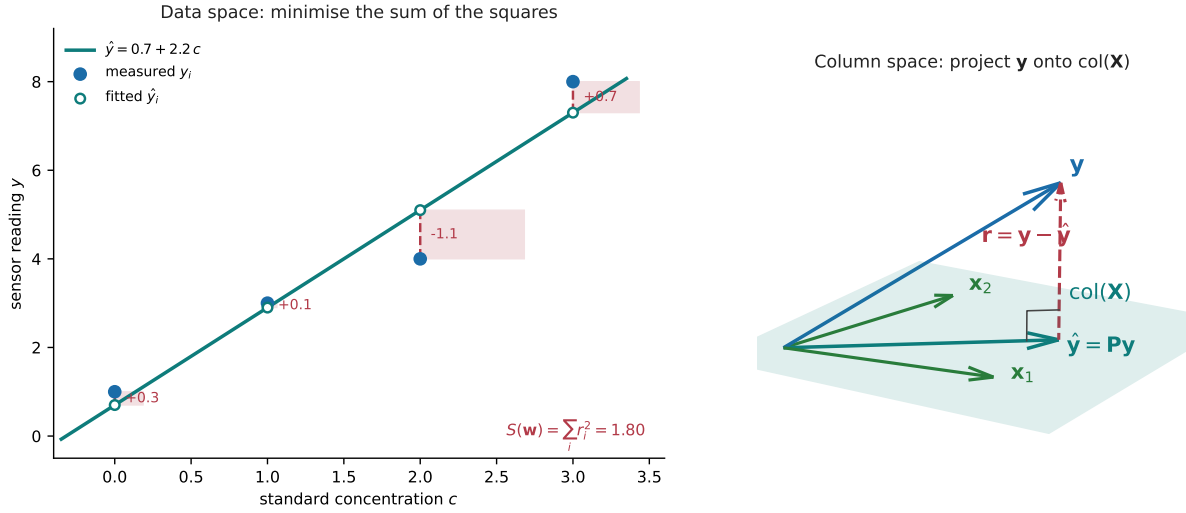


Figure 2: Two pictures of one computation, for the calibration example. **Left, data space:** each sample contributes a vertical residual, and the shaded rectangle beside it has area r_i^2 ; least squares minimises the total shaded area, here 1.80. **Right, column space:** the same four numbers are a single point $\mathbf{y} \in \mathbb{R}^4$, and $\text{col}(\mathbf{X})$ is the plane spanned by the two feature columns. The fit $\hat{\mathbf{y}}$ is the shadow of $\mathbf{y}$ on that plane and the residual meets it at a right angle. The right-hand picture is drawn schematically — the true ambient space here is $\mathbb{R}^4$ — but the perpendicularity it shows is exact.

Two spaces, and the commonest confusion in this topic

Figure 2 shows two different spaces and it is worth being explicit about which is which, because students routinely fuse them.

On the left, each *point* is a sample and the axes are the feature and the response. There are four points, and the space is two-dimensional because there is one feature plus one response.

On the right, each *axis* is a sample and each arrow is a column of the data. There is one point $\mathbf{y}$, and the space is four-dimensional because there are four samples. The vectors $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are the two feature columns, each a list of four numbers.

The residual is vertical on the left and perpendicular to a plane on the right. Both statements are true, of the same four numbers, seen along different axes.

5 The bridge: why the two derivations had to agree

That two such different arguments produce the same equation can look like a coincidence. It is not, and the reason is contained in one line we have already written. Recall (3.8):

$$\frac{\partial S}{\partial w_j} = -2 \langle \mathbf{x}^{(j)}, \mathbf{r} \rangle.$$

Read the two derivations against it.

The calculus route demands $\partial S / \partial w_j = 0$ for every j . The geometric route demands $\langle \mathbf{x}^{(j)}, \mathbf{r} \rangle = 0$ for every j . By (3.8) these differ by the factor -2 . They are not two conditions that happen to have the same solution; they are the *same n equations*, written once in the vocabulary of derivatives and once in the vocabulary of angles.

Key idea

One sentence, two languages. For a squared- L_2 objective, the partial derivative with respect to a coefficient *is* the scalar product of the corresponding feature column with the residual.

Hence “the gradient vanishes” and “the residual is orthogonal to every feature” say the same thing. **Calculus and geometry are not two routes to one answer here; they are one route described twice.**

It is worth seeing why this is special to the squared L_2 norm rather than a general fact. Differentiating $S(\mathbf{w}) = \langle \mathbf{r}, \mathbf{r} \rangle$ by the product rule,

$$dS = 2 \langle d\mathbf{r}, \mathbf{r} \rangle = 2 \langle -\mathbf{X} d\mathbf{w}, \mathbf{r} \rangle = -2 \langle d\mathbf{w}, \mathbf{X}^\top \mathbf{r} \rangle, \quad (5.1)$$

so the gradient of S is $-2\mathbf{X}^\top \mathbf{r}$, a vector of scalar products, because differentiating a squared norm hands you back the vector itself paired through the very scalar product that defined the norm. Requiring that pairing to vanish against every direction $d\mathbf{w}$ is orthogonality, by definition.

Choose L_1 instead and the chain breaks in a specific place. The objective is still minimisable and the subgradient condition still characterises the optimum, but the derivative is no longer a scalar product with the residual — it is a sum of signs, $\partial/\partial w_j \sum_i |r_i| = -\sum_i X_{ij} \operatorname{sgn}(r_i)$ — so there is no orthogonality statement, no projection, and no matrix $\mathbf{P}$ in the next section. The geometry is a privilege of the scalar product, exactly as Module 2, §6.3 warned.

6 The projector made explicit

The geometric derivation says $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$, but so far “projection” has been a description. Substituting (3.5) back into $\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{w}}$ turns it into a matrix:

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{P}\mathbf{y}, \quad \mathbf{P} := \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top. \quad (6.1)$$

Definition

$\mathbf{P} \in \mathbb{R}^{m \times m}$ is the **hat matrix**, so called because it puts the hat on $\mathbf{y}$. It depends only on $\mathbf{X}$ — on *where you measured*, never on *what you measured*. The same $\mathbf{P}$ serves for any response vector collected on the same design.

Three properties identify it as an orthogonal projector, and each is one line.

Symmetry. $\mathbf{P}^\top = (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{P}$, since $\mathbf{X}^\top \mathbf{X}$ is symmetric and so is its inverse.

Idempotence. Projecting twice is the same as projecting once:

$$\mathbf{P}^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \underbrace{\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}}_{=\mathbf{I}} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{P}. \quad (6.2)$$

This is geometrically obvious — a shadow already lying on the plane is its own shadow — and it is the algebraic signature of a projector.

Spectrum and rank. If $\mathbf{P}\mathbf{v} = \lambda\mathbf{v}$ then $\lambda\mathbf{v} = \mathbf{P}\mathbf{v} = \mathbf{P}^2\mathbf{v} = \lambda^2\mathbf{v}$, so $\lambda^2 = \lambda$ and every eigenvalue is 0 or 1. The eigenvectors with $\lambda = 1$ are exactly the vectors already in $\operatorname{col}(\mathbf{X})$, which they are left untouched; those with $\lambda = 0$ are the vectors orthogonal to it, which are annihilated. Hence

$$\operatorname{rank}(\mathbf{P}) = \operatorname{tr}(\mathbf{P}) = n, \quad (6.3)$$

the number of independent features — and the quantity a statistician calls the **degrees of freedom** of the fit.

The complementary matrix $\mathbf{M} = \mathbf{I} - \mathbf{P}$ is also symmetric and idempotent; it projects onto the orthogonal complement $\text{col}(\mathbf{X})^\perp = \ker(\mathbf{X}^\top)$, and it produces the residual:

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{M}\mathbf{y}, \quad \mathbf{P}\mathbf{M} = \mathbf{P} - \mathbf{P}^2 = \mathbf{0}. \quad (6.4)$$

The last identity restates the orthogonality of fit and residual in operator form: the two projectors annihilate each other because their target subspaces are perpendicular.

The projector for the calibration data

For the four-point design of Section 1,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, \quad \det = 20, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} 16 \\ 35 \end{pmatrix},$$

so $\hat{\mathbf{w}} = \frac{1}{20}(14 \cdot 16 - 6 \cdot 35, -6 \cdot 16 + 4 \cdot 35) = (0.7, 2.2)$ and the calibration line is $\hat{y} = 0.7 + 2.2c$. The fitted values and residuals are

$$\hat{\mathbf{y}} = (0.7, 2.9, 5.1, 7.3), \quad \mathbf{r} = (0.3, 0.1, -1.1, 0.7), \quad S = 1.80.$$

Check the orthogonality directly: $\langle \mathbf{x}^{(1)}, \mathbf{r} \rangle = 0.3 + 0.1 - 1.1 + 0.7 = 0$ and $\langle \mathbf{x}^{(2)}, \mathbf{r} \rangle = 0 \cdot 0.3 + 1 \cdot 0.1 + 2 \cdot (-1.1) + 3 \cdot 0.7 = 0$. The first of these is the familiar statement that least-squares residuals sum to zero — which we now see is not a separate fact but simply orthogonality to the intercept column of ones.

The hat matrix is

$$\mathbf{P} = \frac{1}{10} \begin{pmatrix} 7 & 4 & 1 & -2 \\ 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ -2 & 1 & 4 & 7 \end{pmatrix},$$

symmetric, idempotent, with $\text{tr}(\mathbf{P}) = \frac{1}{10}(7 + 3 + 3 + 7) = 2 = n$ and eigenvalues $\{0, 0, 1, 1\}$ — two directions kept, two discarded, as it must be for a projection of $\mathbb{R}^4$ onto a plane.

7 Pythagoras, and where R^2 comes from

Because $\hat{\mathbf{y}}$ and $\mathbf{r}$ are orthogonal, the decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{r}$ is a right triangle, and

$$\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\mathbf{r}\|^2. \quad (7.1)$$

When the model includes an intercept — a column of ones — the constant vector $\bar{y}\mathbf{1}$ lies in $\text{col}(\mathbf{X})$, so the same argument applies after subtracting it, giving the decomposition every statistics course states without proof:

$$\underbrace{\|\mathbf{y} - \bar{y}\mathbf{1}\|^2}_{\text{TSS}} = \underbrace{\|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2}_{\text{ESS}} + \underbrace{\|\mathbf{r}\|^2}_{\text{RSS}}, \quad R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}. \quad (7.2)$$

For the calibration data, $\text{TSS} = 26$, $\text{ESS} = 24.2$, $\text{RSS} = 1.8$ and $R^2 = 0.931$.

R^2 is a cosine

Since $\mathbf{y} - \bar{y}\mathbf{1}$ is the hypotenuse and $\hat{\mathbf{y}} - \bar{y}\mathbf{1}$ the adjacent side of a right triangle, their cosine is $\cos \theta = \|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\| / \|\mathbf{y} - \bar{y}\mathbf{1}\|$, so

$$R^2 = \cos^2 \theta.$$

The coefficient of determination is the squared cosine of the angle between the centred data and the centred fit — the cosine similarity of Module 2, §6.4, arriving in statistics under

an alias. For the calibration example $\theta = 15.3^\circ$, and $\cos^2 15.3^\circ = 0.931$ as computed above. “93% of the variance explained” means, literally, that the fit sits 15° away from the data.

8 One column: the formula you already had

Take $n = 1$, so $\mathbf{X}$ is a single column $\mathbf{x} \in \mathbb{R}^m$ and $\text{col}(\mathbf{X})$ is a line. Then $\mathbf{X}^\top \mathbf{X} = \mathbf{x}^\top \mathbf{x}$ is a scalar and (3.5) reads

$$\hat{w} = \frac{\mathbf{x}^\top \mathbf{y}}{\mathbf{x}^\top \mathbf{x}}, \quad \hat{\mathbf{y}} = \hat{w} \mathbf{x} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} \mathbf{x}, \quad (8.1)$$

which is character for character the projection of one vector onto another from Module 2, §6.5. The hat matrix becomes $\mathbf{P} = \mathbf{x}\mathbf{x}^\top / (\mathbf{x}^\top \mathbf{x})$, of rank one and trace one.

Key idea

The one-line summary of this annex. Least squares is the projection formula, promoted from a line to an n -dimensional subspace. One column is a line; n columns are a subspace; the operator is the same, and $(\mathbf{X}^\top \mathbf{X})^{-1}$ is doing nothing more exotic than what division by $\langle \mathbf{x}, \mathbf{x} \rangle$ did in the scalar case — correcting for the fact that the spanning vectors are neither unit length nor mutually orthogonal.

This last observation is worth pressing. If the columns of $\mathbf{X}$ happened to be orthonormal, $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ and (3.5) would collapse to $\hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y}$ — just take scalar products with each column, no inversion at all. The Gram matrix $\mathbf{X}^\top \mathbf{X}$ is precisely the Gram matrix of Module 2, §6.1, and the whole difficulty of least squares is that in real data it is not the identity. This is also what Gram–Schmidt (and its industrial form, the QR factorisation) is for: orthonormalise the columns first, and the problem becomes trivial.

9 When $\mathbf{X}^\top \mathbf{X}$ is not invertible

Everything above assumed $\text{rank}(\mathbf{X}) = n$. Suppose it is not — two features are perfectly collinear, say, or there are more features than samples. Then $\mathbf{X}^\top \mathbf{X}$ is singular and (3.5) is meaningless. What survives?

The normal equations (3.4) themselves are always consistent: whatever $\mathbf{X}$ is, $\mathbf{X}^\top \mathbf{y}$ lies in $\text{col}(\mathbf{X}^\top \mathbf{X}) = \text{col}(\mathbf{X}^\top)$, so by Rouché–Frobenius a solution exists. There are now infinitely many — add to $\hat{\mathbf{w}}$ any vector in $\ker(\mathbf{X})$ and the equations still hold.

But this is exactly where the two viewpoints of this annex separate to useful effect.

Key idea

The fit is unique; the coefficients are not. The geometric derivation asked for the closest *point* $\hat{\mathbf{y}}$ in $\text{col}(\mathbf{X})$, and the closest-point theorem proved that point unique — rank had nothing to do with it. Rank deficiency does not make the projection ambiguous. It makes the *coordinates* of the projection ambiguous, because a linearly dependent spanning set assigns more than one coordinate list to the same vector. So $\hat{\mathbf{y}}$ and the residuals are always well defined and reproducible. $\hat{\mathbf{w}}$ need not be, and interpreting individual coefficients in the presence of collinearity is therefore unsafe even when the predictions are perfect.

Two standard repairs, both already in Module 2, §14.1:

Pick the smallest solution. The Moore–Penrose pseudoinverse $\mathbf{X}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top$, built from

the SVD, returns the minimum-norm member of the solution set:

$$\hat{\mathbf{w}} = \mathbf{X}^+ \mathbf{y} = \arg \min \{ \|\mathbf{w}\| : \mathbf{w} \text{ minimises } S \}. \quad (9.1)$$

This is what `np.linalg.lstsq` and Scikit-Learn’s `LinearRegression` compute; neither fails on rank-deficient input, they simply answer a slightly sharper question.

Change the problem. Ridge regression adds a penalty,

$$S_\alpha(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \alpha\|\mathbf{w}\|^2 \implies \hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (9.2)$$

which the derivation of Section 3 reproduces with one extra term in the gradient. Read through the spectral theorem, adding $\alpha \mathbf{I}$ shifts every eigenvalue of $\mathbf{X}^\top \mathbf{X}$ from λ_i to $\lambda_i + \alpha$: strictly positive whatever the data did, hence always invertible, and better conditioned. Note that (9.2) is no longer a projection — $\hat{\mathbf{y}}$ is pulled off $\text{col}(\mathbf{X})$ ’s nearest point toward the origin — which is precisely the bias we accept in exchange for variance.

Never form $\mathbf{X}^\top \mathbf{X}$ numerically

Equation (3.5) is the right way to *understand* least squares and the wrong way to *compute* it. The condition number of the product is the square of the original,

$$\kappa_2(\mathbf{X}^\top \mathbf{X}) = \kappa_2(\mathbf{X})^2,$$

so forming it doubles the number of significant digits you lose. A matrix with $\kappa_2(\mathbf{X}) = 10^6$ — unremarkable for real design matrices — gives $\kappa_2(\mathbf{X}^\top \mathbf{X}) = 10^{12}$, which in double precision leaves almost nothing. Production solvers therefore work from the QR factorisation or the SVD of $\mathbf{X}$ directly and never build the product at all. Use `np.linalg.lstsq`, not `np.linalg.inv(X.T @ X) @ X.T @ y`.

10 Summary

Object	Meaning
$\mathbf{X}\mathbf{w} = \mathbf{y}$	Overdetermined, inconsistent whenever $\mathbf{y} \notin \text{col}(\mathbf{X})$, which is always.
$S(\mathbf{w}) = \ \mathbf{y} - \mathbf{X}\mathbf{w}\ ^2$	What we minimise instead. Convex, quadratic, differentiable.
$\mathbf{X}^\top \mathbf{X} \hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y}$	The normal equations: simultaneously “ $\nabla S = \mathbf{0}$ ” and “ $\mathbf{r} \perp \text{col}(\mathbf{X})$ ”.
$\partial S / \partial w_j = -2\langle \mathbf{x}^{(j)}, \mathbf{r} \rangle$	The bridge. Derivative and scalar product are the same number.
$\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$	Orthogonal projector onto $\text{col}(\mathbf{X})$: symmetric, idempotent, $\text{tr} = \text{rank} = n$.
$\ \mathbf{y}\ ^2 = \ \hat{\mathbf{y}}\ ^2 + \ \mathbf{r}\ ^2$	Pythagoras; centred, it is $\text{TSS} = \text{ESS} + \text{RSS}$ and $R^2 = \cos^2 \theta$.
$\mathbf{X}^+ = \mathbf{V} \Sigma^+ \mathbf{U}^\top$	Survives rank deficiency by returning the minimum-norm solution.

Exercises

1. Verify (1.2) for the calibration data by row-reducing $[\mathbf{X} \mid \mathbf{y}]$ and confirming its rank is 3 while $\text{rank}(\mathbf{X}) = 2$.
2. Fit the same four points by hand using (3.5) and confirm $\hat{\mathbf{w}} = (0.7, 2.2)$. Then compute

- $\mathbf{r}$ and check both orthogonality conditions numerically.
3. Show that if $\mathbf{X}$ contains a column of ones, the least-squares residuals always sum to zero. Which orthogonality condition is this?
 4. Prove that $\mathbf{M} = \mathbf{I} - \mathbf{P}$ is symmetric and idempotent, and that $\text{tr}(\mathbf{M}) = m - n$. Why is that number the “residual degrees of freedom”?
 5. Take $\mathbf{P}$ from the worked example and verify $\mathbf{P}^2 = \mathbf{P}$ by hand for its first row. Then confirm $\mathbf{P}\mathbf{x}^{(1)} = \mathbf{x}^{(1)}$ — a vector already in the column space is its own projection.
 6. Add a third column to the calibration design equal to twice the second. Show that $\det(\mathbf{X}^\top \mathbf{X}) = 0$, that $\hat{\mathbf{y}}$ is unchanged, and that `np.linalg.pinv` returns a different $\hat{\mathbf{w}}$ from any you construct by hand — with smaller norm.
 7. Derive the ridge solution (9.2) by repeating Section 3 with the extra penalty term. Then show its hat matrix $\mathbf{P}_\alpha = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^\top$ is symmetric but *not* idempotent for $\alpha > 0$. What does that tell you geometrically?
 8. Suppose the columns of $\mathbf{X}$ are orthonormal. Show $\hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y}$ and $\mathbf{P} = \mathbf{X} \mathbf{X}^\top$, and explain why Gram–Schmidt on the columns is a legitimate — if wasteful — way to solve any least-squares problem.

Where this goes next. The pseudoinverse of (9.1) is built from the SVD (Module 2, §13), and the same projector reappears in PCA (§14.2), where the subspace being projected onto is chosen to maximise retained variance instead of being handed to you as $\text{col}(\mathbf{X})$. In Géron, Chapter 4 computes (3.5) directly, then abandons it for gradient descent — which is worth understanding as a numerical retreat, not a conceptual one: the closed form is still correct, it is merely $O(n^3)$ and numerically delicate, and for large n an iterative walk downhill on the bowl of Figure 1 is cheaper than inverting anything.